SENIOR HIGH SCHOOL

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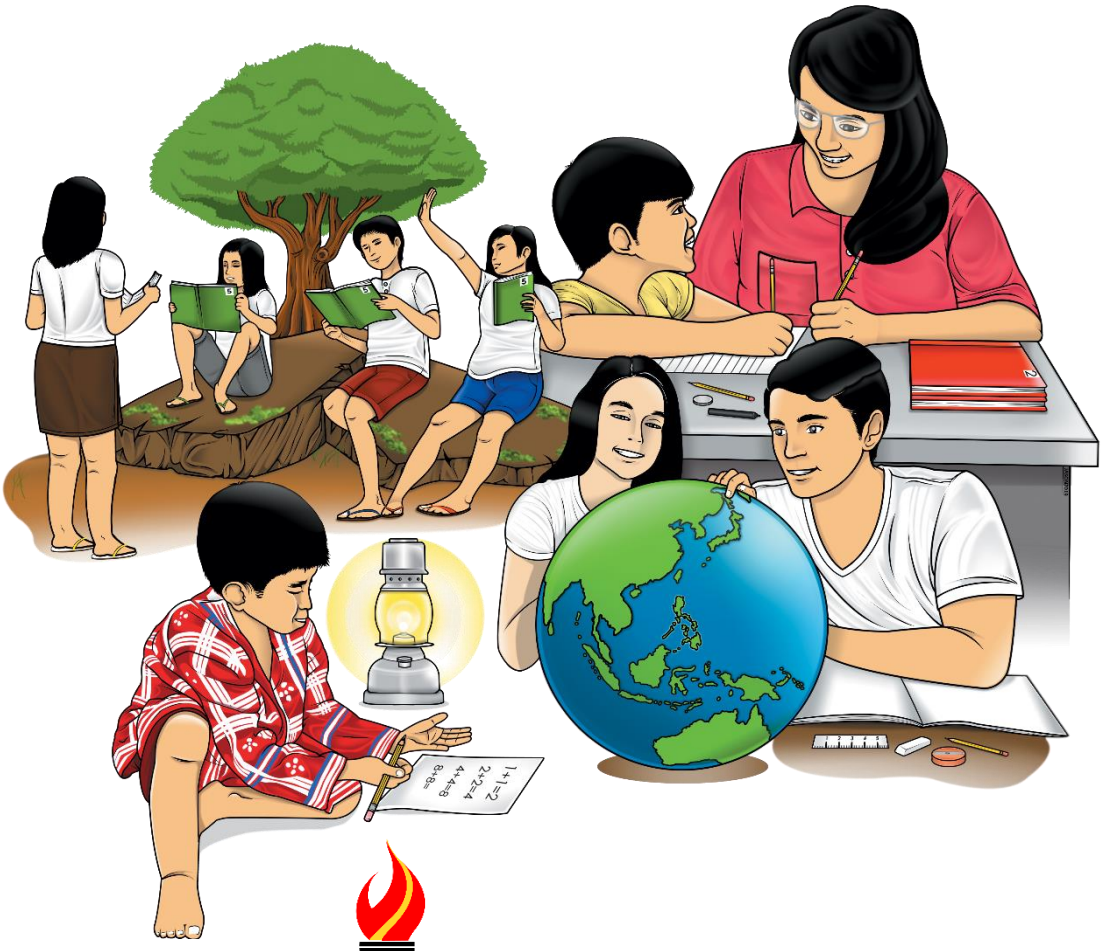
DEPARTMENT OF EDUCATION

GENERAL MATHEMATICS
Quarter 2



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What I Need to Know

Targets:

1. Illustrate simple and compound interests (**M11GM-IIa-1**);
2. Distinguish between simple and compound interests (**M11GM-IIa-2**);
3. Compute interest, maturity value, future value, and present value in simple interest **(M11GM-IIa-b-1)**; and
4. Solve problems involving simple interest **(M11GM-IIb-2)**.

What I Know

Write the letter of the correct answer.

1. **Leoncio borrowed ₱150,000.00 from a loan firm with a rate of simple interest of 15% per annum. How much will Leoncio has to pay at the end of the year?**
   A. ₱ 172,500.00 B. ₱ 127.000.00 C. ₱ 157,000.00 D. ₱ 175,000.00
2. **If ₱ 3,000.00 accumulates to ₱ 5,010.00 after four years when invested at simple interest, what is the interest rate?**
   A. 17.65% B. 15.75% C. 16.75% D. 17.65%
3. **If ₱ 80,000.00 was borrowed from ABC bank with 12% simple interest rate annually, what should be the amount to be paid after 10 months?**
   A. ₱87,000.00 B. ₱ 89,000.00 C. ₱ 92,000.00 D. ₱ 88,000.00
4. **Cassie invested ₱ 60,000.00 in XYZ financing institution at 7% simple interest rate per annum. How long Cassie will have to wait to reach her money to ₱ 99,900.00?**
   A. 7 years B. 9 ½ years C. 8 years D. 10 ½ years
5. **How much should be invested to reach an amount of ₱ 50,000.00 at 5% simple interest for 5 years?**
   A. ₱ 30,500.00 B. ₱ 35,000.00 C. ₱ 40,000.00 D. ₱ 45,000.00

Lesson 1 Simple Interest

What's New

Interest. It pertains to the payment from borrowing a principal amount of money over a period of time.

Simple interest. It refers to the amount calculated by multiplying the principal, interest rate and a period of time.

Principal amount. This is the initial amount or capital money borrowed or loaned from a financial institution.

Interest rate. This is the percentage grow of capital value.

Time. It is the length of term or period from which the principal amount was borrowed.

Future Value. It is the sum of the principal amount and interest.

What is It

Simple Interest Formula	Future Value Formula
$I = Prt$	$F = P + I$
I – simple interest	F – future or maturity value
P – principal or capital amount	I – interest
r – interest rate	
t – time in years	

Note that from

$F = P + I$ → Future Value formula

$F = P + Prt$ → Substitute I with Prt from Simple Interest formula

$F = P(1 + rt)$ → factor-out common variable P

Thus, Future Value formula can also be $F = P(1+rt)$.

What's More

	Calculation
1. Find the simple interest of ₱6,000.00 at 2% for 3 years.	$I = Prt$ → formula $I = (₱6{,}000.00)(0.02)(3)$ →substitute given given $I = ₱\ 360.00$
2. Find the future value of ₱ 520,000.00 at 10% for 2 years.	$F = P(1+rt)$ → formula $F = (₱520{,}000)(1+((0.10)(2))$→substitute given $F = (₱520{,}000)(1+0.20)$ →simplify $F = (₱520{,}000)(1.20)$ →simplify $F = ₱\ 624{,}000.00$
3. How long will ₱ 10,000.00 take to reach ₱ 25,000.00 at 5% simple interest rate.	$F = P(1+rt)$ → formula $₱25{,}000 = ₱10{,}000(1+0.05t)$ →substitute given $[₱25{,}000 = ₱10{,}000(1+0.05t)]\frac{1}{₱10{,}000}$ →multiply the whole equation by multiplicative inverse of ₱10,000.

	2.5 = 1 + 0.05t → simplify 2.5 – 1 = 0.05t → by additive inverse 1.5 = 0.05t → simplify $[1.5 = 0.05t]\frac{1}{0.05}$ → multiply the whole equation by multiplicative inverse of 0.05 t = 30 ∴ It takes 30 years.
4. Determine the present value of a loan with future amount of ₱142,000.00 at 7% simple interest rate for 6 years.	F = P(1+rt) → formula ₱142,000=P(1+ (0.07 • 6)) →substitute given ₱142,000 = P(1 + 0.42) → simplify ₱142,000 = P(1.42) → simplify $[₱142,000=P(1.42)]\frac{1}{1.42}$ →multiply the whole equation by Multiplicative inverse of 1.42 ₱100,000 = P ∴ The Principal amount is ₱100,000.00

What I Can Do

Ruth borrowed ₱200,000.00 with 20% simple interest rate from ABC bank. At the time the money was borrowed, the interest was deducted. After a year, Ruth has to pay ₱200,000.00. What is the actual interest rate?

Assessment

1) Compute the simple interest of ₱125,000.00 at 4% for 18 months.
2) Compute the future value of ₱84,000.00 at 8 ½ % for 5 years.
3) Find the simple interest rate of an initial amount of ₱76,000.00 for 3 years with an interest of ₱5,700.00.
4) If you are to lend money with simple interest, one with ₱146,050.00 at 3.7% for 8 years and the other one with ₱164,400.00 at 2.8% for 9 years, which one will earn more?
5) How long will it take ₱6,000.00 to double at 10% simple interest?

What I Need to Know

Targets:

1. Illustrate simple and compound interests (**M11GM-IIa-1**).
2. Solve problems involving compound interest **(M11GM-IIb-2)**.

What I Know

Write the letter of the correct answer.

1. **John borrowed PHP 153, 000 at 8% compounded annually. How much shall he pay after 3 years?**
   A. ₱129, 375.49 C. ₱195, 372.94
   B. ₱192, 735.94 D. ₱197, 273.94
2. **Suppose you invested PHP 35, 000 at a simple rate of 2.5%, how much will be your investment after 10 years?**
   A. ₱73.872.77 C. ₱83,728.80
   B. ₱73,873.77 D. ₱83,872.00
3. **What amount must be invested in order to have ₱61,700.08 after 5 years if money is worth 5% compounded semi- annually?**
   A. ₱60,657.08 C. ₱61,657.91
   B. ₱60,700.08 D. ₱61,700.08

Lesson 2 Compound Interest

What's New

Compound interest (I_C) is the addition of **interest** to the principal sum of a loan or deposit, or in other words, **interest** on **interest**. It is the result of reinvesting **interest** rather than paying it out. Interest is computed on the principal and also on accumulated past interest. In **compound interest**, the interest **from the previous year also earns interest**. Thus, the interest **grows every year.**

What is It

MATURITY (FUTURE) VALUE

$$F = P(1+r)^t$$

where:

F - maturity (future) value
P - principal, present value
r – interest rate
t - time in years

COMPOUND INTEREST

$$I_c = F - P$$

where: I_c – compound interest

PRESENT VALUE

$$P = \frac{F}{(1+r)^t}$$

MATURITY VALUE (COMPOUNDING)

$$F = P(1+j)^n$$

$$F = P(1+\frac{i^m}{m})^{tm}$$

where:

i^m - nominal rate (annual)
m - frequency of conversion
$j = i/m, \quad n = tm$
j - interest rate per period
n - total number of conversion

ILLUSTRATION OF SIMPLE AND COMPOUND INTEREST

Find the maturity value if ₱20, 000 is invested in a bank at 3% compounded annually for 5 years.

Time (t)	**Principal (P)**	**Interest rate (r)**	**Simple Interest**		**Amount after t years (Maturity Value)**
			Solution	Answer	
1	₱20, 000	3%	(₱20, 000)(0.03)(1)	₱600	₱20, 000 + ₱600 = ₱20, 600
2	₱20, 600		(₱20, 600)(0.03)(1)	₱618	₱20, 600 + ₱618 = ₱21, 218
3	₱21, 218		(₱21, 218)(0.03)(1)	₱636.54	₱21, 218 + ₱636.54 = ₱21, 854.54
4	₱21, 854.54		(₱21, 854.54)(0.03)(1)	₱655.64	₱21, 854.54 + ₱655.64 = ₱22, 510.18
5	₱22, 510.18		(₱22, 510.18)(0.03)(1)	₱675.31	₱22, 510.18 + ₱675.31 = ₱23, 185.49

Illustration1: Using Simple Interest (Annual)

What's More

Year (t)	Principal, P = ₱20, 000 ; Interest rate, r = 3%	
	Amount of money at the end of the year	
1	$P\ x\ (1+r) = \boldsymbol{P(1+r)}$	₱20, 000(1.03) = **₱20, 600**
2	$\boldsymbol{P(1+r)\ x}\ (1+r) = \boldsymbol{P(1+r)^2}$	**₱20, 000**$(1.03)^2$ **= ₱21, 218**
3	$\boldsymbol{P(1+r)^2}\ x\ (1+r) = \boldsymbol{P(1+r)^3}$	**₱20, 000**$(1.03)^3$ **= ₱21, 854.54**
4	$\boldsymbol{P(1+r)^2}\ x\ (1+r) = \boldsymbol{P(1+r)^3}$	**₱20, 000**$(1.03)^4$ **= ₱22, 510.18**
5	$\boldsymbol{P(1+r)^2}\ x\ (1+r) = \boldsymbol{P(1+r)^3}$	**₱20, 000**$(1.03)^5$ **= ₱23, 185.49**

Illustration 2: Using Compound Interest

What I Can Do

A. Complete the table.

Time (t)	**Principal (P)**	**Interest rate (r)**	**Simple Interest**		**Amount after t years (Maturity Value)**
			Solution	Answer	
1	₱18, 000	5%			
2					
3					
4					
5					

B. Answer the following word problems.

1) Claude deposited ₱65, 700.00 in a savings account which pays 8% interest compounded semi – annually. What will his balance be in 5 years?
2) Find the compound amount due in 10 years if ₱400, 000.00 is invested at 10% compounded monthly.
3) If Natan wants ₱5, 000.00 three years from now and the compounded interest rate is 7%, how much should he invest today?

Assessment

Solve the following problems on compound interest.

1) Miya deposits ₱5, 000.00 in a savings account which pays 8% compounded annually. The money is left for 4 years. Find the account balance at the end of 4 years.
2) What is the interest and maturity value of a loan for ₱30, 000 at 8% compound interest for 4 years?
3) For Roger to have ₱60, 000 in 6 years, how much should he invest if the compound interest is 6%?

What I Need to Know

Targets:

1. Illustrate simple and general annuities (**M11GM-IIc-1**).
2. Distinguish between simple and general annuities (**M11GM-IIc-2**).
3. Find the future value and present value of both simple annuities and general annuities (**M11GM-IIc-d-1**).

What I Know

Write the letter of the correct answer.

1. What will be the future value at the end of the last payment for a ₱500.00 amount per payment made every year for 3 years with 4% compounded annually?

A. ₱1, 650.80 B. ₱1, 560.80 C. ₱1, 850.60 D. ₱1, 680.50

2. Jackie plans to invest money in a financial institution. What will be his deposit per year in the financial institution that gives 3% interest compounded annually to accumulate ₱300 000.00 at the end of 10 years?

A. ₱26, 169.152 B. ₱ 26, 916.152 C. ₱ 29, 615. 162 D. ₱ 29, 652.192

Lesson 3 General Annuities

What's New

Annuity. It refers to the series of same amount of payment at regular intervals.
Term of an annuity (t). It is the time between the first payment interval and last payment interval.
Regular or Periodic payment (R). It refers to the amount of each payment.
Amount (Future Value) of an annuity (F). This is the sum of the future values of all the payments to be made during the entire term of the annuity.
Present Value of an annuity (P). It is the sum of the present values of all the payments to be made during the entire term of the annuity.

What is It

Future Value of ordinary annuity

$$F = R\,\frac{(1+j)^n - 1}{j}$$

Present Value of ordinary annuity

$$P = R\,\frac{1-(1+j)^{-n}}{j}$$

where: R is the regular payment
j - interest rate per period
n - number of payments

Source: General Mathematics (Learning Module, DepEd, p. 170)

What's More

1) Cally deposits ₱5,000.00 every year in a bank that gives 6.85% interest compounded annually. How much will be in his account at the end of 6 years?

This is an example of future value of ordinary annuity hence we need to compute the sum of money in the entire term.

r = ₱5,000.00

j = 6.85% or 0.0685

n = 6

$$F = R\,\frac{(1+j)^n - 1}{j}$$

$$F = ₱5,000\,\frac{(1+0.0685)^6 - 1}{0.0685}$$

F = ₱35, 631.49949

F = ₱35, 631.50 (round off to the nearest hundredths)

2) Determine the future value and present value of an ordinary annuity for ₱500,000.00 investment at 4 % interest compounded quarterly for 5 years.

$$R = \frac{₱500\,00.00}{4} = ₱125\,000.00$$

$$j = \frac{4\%}{4} = \frac{0.04}{4} = 0.01$$

$$n = 5(4) = 20$$

The future value of ordinary annuity

$$F = R\,\frac{(1+j)^n - 1}{j}$$

$$F = ₱125\,000\,\frac{(1+0.01)^{20} - 1}{0.01}$$

F = ₱ 2 752 375.499

F = ₱ 2 752 375.50 (round off to the nearest hundredths)

The future value of ordinary annuity

$$P = R \cdot \frac{1 - (1+j)^{-n}}{j}$$

$$P = ₱125,000 \cdot \frac{1 - (1+0.01)^{-20}}{0.01}$$

P = ₱ 2,255,694.121

P = ₱ 2,255,694.12 (round off to the nearest hundredths)

What I Can Do

Find the present value (P) and future value (F) of the ordinary annuity:

1) Quarterly payments of P2,000 for 5 years with interest rate of 8% compounded quarterly

2) Semi-annual payment of P8,000 for 12 years with an interest rate of 12% compounded semi-annually

Assessment

Solve and answer the following problems.

1) Peter started to deposit P5,000 annually in a fund that pays 1% interest rate compounded quarterly. How much will the fund be after 6 years?
2) The buyer of the lot pays P50,000 down payment cash, and P10,000 every month for 10 years. If money is 8% compounded monthly, how much is the cash value of the lot?
3) How much should be invested in a fund each year, paying 2% compounded annually to accumulate P100,000 in 5 years?

What I Need to Know

Targets:

1. Calculate the fair market value of a cash flow stream that includes an annuity (**M11GM-IId-2**).
2. Calculate the present value and period of deferral of a deferred annuity (**M11GM-IId-3**).

What I Know

Choose the letter of the letter of the correct answer.

1. **Air Condition Unit is for sale at P36, 000 in cash or monthly of P6,360 for 6 months at 12% compounded semi–annually. Which is lower, the cash price or the present value of the installment term?**

   A. P36,889.80 B. P38,160.00 C. P40,579.20 D. P38,689.00

Lesson 4 Fair Market Value and Deferred Annuities

What is It

FAIR MARKET VALUE (FMV)

The **Fair Market Value** or **economic value** of a cash flow (payment stream) on a particular date refers to a *single amount* that is equivalent to the value of the payment stream at that date. This specific date is called the **focal date**.

A **cash flow** is a term that refers to payments received (cash inflows) or payments or deposits made (cash outflows). Cash inflows can be represented by positive numbers and cash outflows can be represented by negative numbers.

DEFERRED ANNUITY

Deferred Annuity. This is an annuity that does not begin until a given time interval has passed.

Period of Deferral. It is the time purchase of an annuity and the start for the deferred annuity.

What's More

Mr. Johnson received two offers on a lot that he wants to sell. Mr. Kennedy has offered P80, 000 and a P1, 000, 000 lump sum payment 5 years from now. Mr. Francisco has offered P80, 000 plus P40, 000 every quarter for 5 years. Compare the fair market values of two offers if money can earn 5% compounded annually. Which offer has a higher market value?

Choose a focal date and determine the values of two offers at that focal date. For example, the focal date can be at the start of the term.

Since the focal date is at t = 0, compute for the present value of each offer.

Mr. Kennedy's offer P80, 000 down payment P1, 000, 000 after 5 years **PRESENT VALUE** $P = F(1+j)^{-n}$ $\boldsymbol{P = P1,000,000\,(1+0.05)^{-5}}$ $\boldsymbol{P = P783,526.20 + P80,000}$ $\boldsymbol{P = P863,526.20}$ **Hence, Mr. Kennedy's offer has a higher market value.**	**Mr. Francisco's offer** P80, 000 down payment P40, 000 every quarter for 5 years **SOLVING THE EQUIVALENT RATE Compounded Quarterly of 5% Compounded Annually** $F_1 = F_2$ $P(1+\frac{i^4}{4})^{(4)(5)} = P(1+\frac{i^1}{1})^{(1)(5)}$ $(1+\frac{i^4}{4})^{(20)} = (1+\frac{0.05^1}{1})^{(5)}$ $(1+\frac{i^4}{4})^{(20)} = (1.05)^{(5/20)}$ $\frac{i^4}{4} = (1.05)^{(1/4)} - 1$ $\boldsymbol{\frac{i^4}{4} = 0.0122723443}$ $\boldsymbol{P = R\,\frac{1-(1+j)^{-n}}{j}}$ $\boldsymbol{P = 40,000\,\frac{1-(1+0.0122723443)^{-20}}{0.0122723443}}$ $\boldsymbol{P = P705,571.98}$ $FairMarket\ Value = DP + P$ $FMV = P80,000 + P705,571.98$ $\boldsymbol{FVM = P785,571.98}$

ALTERNATE SOLUTION FUTURE AMOUNT	ALTERNATE SOLUTION FUTURE AMOUNT
$\boldsymbol{F = P(1+j)^n}$ $\boldsymbol{F = P80,000(1+0.05)^5}$ $\boldsymbol{F = P102,102.53}$ $\boldsymbol{FMV = P1,000,000 + P102,102.53}$ $\boldsymbol{FMV = P1,102,102.53}$ **Still, Mr. Kennedy's offer has a higher market value.**	$\boldsymbol{F = R\ \frac{(1+j)^n - 1}{j}}$ $\boldsymbol{F = P4,000\ \frac{(1+0.0122723443)^{20} - 1}{0.0122723443}}$ $\boldsymbol{F = P900,510.37}$ $\boldsymbol{FMV = P102,102.53 + P900,510.37}$ $\boldsymbol{FMV = P1,002,612.90}$

FROM ILLUSTRATION 2: On his 40th birthday, Mr. Richards decided to buy a pension plan for himself. This plan will allow him to claim P20, 000 quarterly for 5 years, starting 3 months after his birthday. What one–time payment should he make on his 40th birthday to pay off this pension plan if the interest rate is 8% compounded quarterly.

Given: **R** = P20, 000 **m** = 4 $\mathbf{i^4}$ = 0.06
Find: **P**

Solution:

The annuity is deferred for 20 years, and it will go on for 5 years. The first payment is due three months (one quarter) after his 60th birthday or at the end of the 81st conversion period. Thus, there are 80 artificial payments.

Number of artificial payments: $\boldsymbol{k = (m)(t) = (4)(20) = 80}$
Number of actual payments: $\boldsymbol{n = (m)(t) = (4)(5) = 20}$
Interest rate per period: $\boldsymbol{j = \frac{i^4}{m} = \frac{0.08}{4} = 0.02}$

The present value P of a deferred annuity is given by the formula:

$$\boldsymbol{P = R\ \frac{1-(1+j)^{-(k+n)}}{j} - R\frac{1-(1+j)^{-k}}{j}}$$

$$\boldsymbol{P} = P20,000\ \frac{1-(\mathbf{1+0.02})^{-(\mathbf{80+20})}}{0.02} - P20,000\frac{1-(\mathbf{1+0.02})^{-\mathbf{80}}}{0.02}$$

$$\boldsymbol{P} = P20,000\ \frac{1-(\mathbf{1.02})^{-(\mathbf{100})}}{0.02} - P20,000\frac{1-(\mathbf{1.02})^{-\mathbf{80}}}{0.02}$$

$$\boldsymbol{P} = P20,000\ \frac{\mathbf{1-0.1380329672}}{0.02} - P20,000\frac{\mathbf{1-0.2051097282}}{0.02}$$

$$\boldsymbol{P} = P20,000\ \frac{\mathbf{0.8619670328}}{\mathbf{0.02}} - P20,000\frac{\mathbf{0.7948902718}}{\mathbf{0.02}}$$

$$\boldsymbol{P = P20,000\ (43.09835164) - P20,000\ (39.74451359)}$$

$$\boldsymbol{P = P861.967.0328 - P794,890.2718}$$

$$\boldsymbol{P = P67,076.761}$$

Therefore, the present value of these monthly pensions is **P67,076.80**

What I Can Do

Answer the word problems and solve for what is being asked.

1. The buyer of a car pays P5 000 every month for 10 years. If money is 8% compounded annually, how much is the cash value of the car?
2. In a sequence of 10 annual payments of P3, 000.00 each, the first one is due at the end of 6 years. Find the present value of deferred annuity if money is worth 8% compounded annually.

Assessment

Answer the word problems and solve for what is being asked.

1. Junnel purchased a cellphone through the credit cooperative of their company. The cooperative provides an option of deferred payment. Junnel decided to pay after 4 months of purchase. His monthly payment is computed as P4, 500 payable in 12 months. How much is the cash value of the cellphone if the interest rate is 6% convertible monthly?

2. A deferred annuity is purchased that will pay P8, 000 per quarter for 20 years after being deferred 10 years and with interest rate of 8 % compounded quarterly. What is the present value of the annuity?

What I Need to Know

Target:
1. Illustrate stocks and bonds. **(M11GM-IIe-1)**
2. Distinguish between stocks and bonds. **(M11GM-IIe-2)**
3. Describe the different markets for stocks and bonds. **(M11GM-IIe-3)**
4. Analyze the different market indices for stocks and bonds. **(M11GM-IIe-4)**

What I Know

Write **true** if the statement is correct. Otherwise, write **false**.

1) Stocks are shares in the ownership of a company.
2) Stocks provide a guaranteed fixed income.
3) Bond is a form of raising money by borrowing from investors.
4) In case of bankruptcy, the shareholder has the first claim on the assets over the bondholder.
5) Stock market index is composed of companies carefully selected to represent the general movement of market prices.

Lesson 5 Basic Concepts of Stocks and Bonds

Suppose you have ₱1,000,000.00, and you want to make it grow. Your friend asked you to become his business partner if you invest your ₱1,000,000.00 in his business. However, a small company offered to borrow your money and pay you back plus the interest at a given time. Where would you invest and why?

What's New

What are STOCKS and BONDS?

Businessmen look for investments where their money may earn. Companies raise funds by borrowing or selling shares to finance their operations or expansions.

https://financialtribune.com/articles/world-economy/24577/philippine-stocks-

Stocks are shares that a company sold to an individual or another company. It is a share in the ownership of the company. Stocks may be bought or sold in a Stock Market. The Philippine Stock Exchange, Inc. (PSE) is the national stock exchange of our country.

So, why do companies sell stocks, and why do people or other companies buy them? Companies sell stocks to gain revenue, to build and grow the company. On the other hand, people buy stocks to invest.

What is It

STOCKS

- A form of **equity financing** or raising money by allowing investors to be part owners of the company
- Stock prices vary every day. These prices are reported in various media (newspaper, TV, internet, etc.)
- Investors can earn if the stock prices increase, but they can lose money if the stock prices decrease, worsen, or if the company goes bankrupt.
- Higher risk but the possibility of higher returns

BONDS

- A form of **debt financing or** raising money by borrowing from investors.
- Investors are guaranteed interest payments and a return of their money at the maturity date
- Investors still need to consider the borrower's credit rating. Bonds issued by the government pose less risk than those by companies because the government has guaranteed funding (taxes) from which it can pay its loans.
- Lower risk but lower yield
- Can be appropriate for retirees (because of the guaranteed fixed

STOCKS	BONDS
• Can be appropriate if the investment is for the long term (10 years or more). It can allow investors to wait for stock prices to increase if ever they go low.	income) or for those who need the money soon (because they cannot afford to take a chance at the stock market) • Bondholders do not vote in the institution's annual meeting but first to claim in the institution's earnings. • In case of bankruptcy, the bondholder has the first claim on the assets over the stockholder.

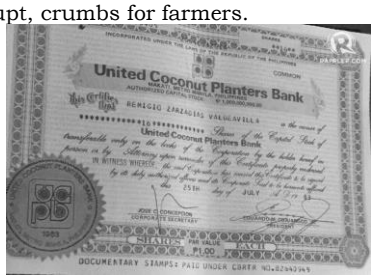
What's More

Imagine this: if you purchase a share of stock in Facebook, that means you own a small portion of that company. You are now considered a **stockholder** or **shareholder** because you own a fraction of the company. Cool, right?

Now, as proof of being a stockholder or shareholder, a **stock certificate** (shown on the right) will be given to you to serve as proof of ownership. Indicated in that stock certificate is the number of stocks or shares you own.

Other companies, however, sell bonds instead of stocks to raise funds. **Bonds** are interest-bearing certificates that show that the owner has loaned money to the bond issuer.

Stock Certificate. The picture of a stock certificate (shown below) is from an article in rappler.com. Read the entire story to know how stock certificates are involved in the lives of the coconut farmers in Quezon and why this article of Camille Elemia says Gold for the corrupt, crumbs for farmers.



https://www.rappler.com/newsbreak/in-depth/158066-coco-levy-fund-scam-coconut-

So, if you choose to lend your money, which means you let the company borrow your money, they will issue you a certificate indicating the amount they have to pay and the maturity date or the date when they must give your money back including the interest. You are then considered a **bondholder**.

Now that you have an idea about stocks and bond, let us focus on each of them.

A. Learn more about STOCKS

If you own stocks in a company; then you will receive **dividends** which is your share in the company's profit. Your dividend is based on **dividend per share,** which is the ratio of the dividends to the number of shares. It means that the dividend you will receive is based on the number of stocks or shares you own. Thus, the larger the stocks you own, the greater the dividends you will get.

- ✓ **Market value** is the current price of a stock at which it can be sold.
- ✓ **Stock Yield Ratio** is the ratio of the annual dividend per share and the market value per share. It is also called current stock yield.

- ✓ **Par Value** is the per share amount as stated on the company certificate. Unlike the market value, par value is determined by the company and remains stable over time.

B. Learn more about BONDS

Unlike the stockholders, bondholders are lenders to the institution, a government, or a private company. Some bond issuers are the national government, government agencies, government-owned and controlled corporations, non – bank corporations, banks, and multilateral agencies.

Comparing Price of a bond P and Par or Face Value F
If P = F, the bond is purchased at par
If P < F, the bond is purchased at a discount
If P > F, the bond is purchased at premium

- ✓ **Bonds** are interest-bearing security that promises to pay (1) a stated amount of money on the maturity date, and (2) regular interest called coupons.
- ✓ **Coupon** is a periodic interest payment that the bondholder receives between purchase date and maturity date, usually semi–annually.
- ✓ **Coupon rate** is the rate per coupon payment period; denoted by *r.*
- ✓ **Price of a bond** is the price of the bond at purchase time; denoted by *P.*
- ✓ **Par Vale of Face Value** is the amount payable on the maturity date; denoted by *F.*
- ✓ **Term of a bond** is the fixed period, when the bond is redeemable, as stated in the bond certificate. It is the number of years from the time of purchase to the maturity date.
- ✓ **Fair price of a bond** is the present value of all cash inflows to the bondholder.

What I Can Do

EXAMPLES ON STOCKS: COMPUTING DIVIDENDS

1. A corporation sold 300,000 shares of common stock to stockholders. The company declared a dividend of ₱30,000,000 at the end of the year. How much is the dividend per share of common stocks?

   Remember that **dividend per share** is simply the ratio of the dividends to the number of shares. Thus,

$$Dividend\ per\ share = \frac{Total\ Dividend}{Total\ Shares} = \frac{₱30{,}000{,}000}{300{,}000} = ₱100$$

   Therefore, the dividend is ₱100 per share of stocks.

2. A chocolate company, Sweets 'r Us, declared a 3% dividend on a stock with a par value of ₱500. Being a chocolate lover, you bought 200 shares of stock with a par value of ₱500. How much dividend will you receive?
   Remember that **par value** is simply the per share amount stated on the company certificate. Thus,

Dividend = (Dividend Percentage) x (Par Value) x (Number of shares)
= 3% x ₱500 x 200
= 0.03 x ₱500 x 200
Dividend = ₱ 3,000
Therefore, the dividend you will receive is ₱ 3,000.

EXAMPLES ON STOCKS: COMPUTING CURRENT YIELD

The Current yield or **stock yield ratio** shows how much dividend is earned in the percentage of the current price of stock per share. It helps the investor decide whether to invest or not to invest, or increase the investment or withdraw. It is computed as:

$$Current\ \ yield = \frac{Annual\ Dividend\ per\ share}{Curret\ price\ per\ share} = \frac{Dividend\ per\ share}{market\ value}$$

Example: FLY airlines is currently selling at ₱2,500 per share. If the earnings per share is ₱150, what is the current yield of stock?

$$Current\ \ yield = \frac{Annual\ Dividend\ per\ share}{Curret\ price\ per\ share} = \frac{₱150}{₱2{,}500} = 0.06\ or\ 6\%$$

The current yield = 0.06 or 6%

EXAMPLE ON BONDS

Determine the amount of the semi-annual coupon for a bond with a face value of ₱300,000 that pays 10% payable semi–annually for its coupons

EXAMPLE ON MARKET INDEX

Annual coupon amount = (Face value) x (Coupon rate) = (₱ 300,000) x (0.10) = ₱ 30,000

Semi-annual coupon amount = $\frac{₱30{,}000}{2}$ = ₱15, 000

Therefore, the amount of semi–annual coupon is **₱15,000**

52 weeks							
HI	LO	STOCK	DIV	YLD%	VOL(100s)	CLOSE	NETCHG
120	105	GGG	3.5	2.8	4050	118.50	-0.50
16	12	HHH	0.9	1.1	1070	15.80	0.10

Consider the following listing on stocks and answer the questions that follow:

For stocks GGG and HHH:

	Answers	
	For stock GGG	**For stock HHH**
1) What was the lowest price of the stock for the last 52 weeks?	₱ 105.00	₱ 12

2) What was the dividend per share last year?	₱ 3.50	₱ 0.9
3) What was the annual percentage yield last year?	2.8%	1.1%
4) What was the closing price on the last trading day?	₱ 118.50	₱ 15.80
5) What was the closing price the day before the last trading day?	₱ 118.50 + ₱ 0.50 = **₱ 119.00**	₱ 15.80 - ₱ 0.10 = **₱ 15.70**

Assessment

Answer the following problems.

1. Ken is planning to sell his shares in a company. The dividend he received was ₱85.00 per share, and the stock price per share is ₱1050.00. What is the current yield of stock?
2. A bank declared a dividend of ₱27 per share for the common stock. If the stock closes at ₱93, how large is the stock yield ratio on this investment?
3. Find the amount of the semi–annual coupon for a ₱200,000 bond which pays 5% convertible semi–annually for its coupons.
4. Consider the following listing on stocks and answer the questions that follow:

For stocks RRR and KKK:

52 weeks							
HI	LO	STOCK	DIV	YLD%	VOL(100s)	CLOSE	NETCHG
75	65	RRR	2.5	2.8	1500	70	2
34	23	KKK	1.7	1.75	1200	28	-3

1) What was the lowest price of the stock for the last 52 weeks?
2) What was the dividend per share last year?
3) What was the annual percentage yield last year?
4) What was the closing price on the last trading day?
5) What was the closing price the day before the last trading day?

What I Need to Know

Targets:

1. Illustrate business and consumer loans. **(M11GM-IIf-1)**
2. Distinguish between business and consumer loans. **(M11GM-IIf-2)**
3. Solve problems involving business and consumer loans (amortization, mortgage). **(M11GM-IIf-3)**

What I know

Problem No. 1. A loan of ₱30,000.00 is to be repaid in full after 2 years. If the interest rate is 9% per annum. How much should be paid after 2 years?

Problem No. 2. If a car loan of ₱790,000.00 requires a 20% down payment. How much is the mortgage?

Lesson 6 Basic Concepts of Business and Consumer Loans

Consumer loans are loans given to individuals for personal or family purpose, while business loans are loans for businesses. Consumer and business loans may require collateral. The collateral for both loans may be real estate or other investments. For business loans, they can use equipment, fixtures, or furniture as collateral. Consumer loans do not usually require a guarantor. Business loans require the business owners to sign as guarantors. For consumer loans, the bank or the lending institution may require a credit report, bank statements, and an income tax return. If the lendee is employed, a certificate of employment and employee payslips. For business loans, the lendee has to submit a credit report, income tax returns, and the company's financial statement. The term of the business loan is generally shorter than the consumer loan. The interest rate for the business loan is usually higher than that for the consumer loan.

A mortgage is a business loan or a consumer loan that is secured with collateral. Collaterals are assets that can ensure a loan. If a borrower cannot pay the loan, the lender has a right to the collateral. The most common collaterals are real estate property. For business loans, equipment, furniture, and vehicles may also be used as collateral. Usually, the loan is secured by the property bought. For example, if a house and lot are purchased, the purchased house and lot will be used as a mortgaged property or collateral. During the loan term, the mortgagor, the borrower in a mortgage, still has the right to possess and use the mortgaged property. If the mortgagor does not make regular payments on the mortgage, the mortgagee or the lender in a mortgage can repossess the mortgaged property. The most common type of mortgage is the fixed-rate mortgage, wherein the interest remains constant throughout the loan term.

What's New

Business Loan. It refers to the money lent specifically for a business purpose. It may be used to start a business or to have a business expansion.
Collateral. These are assets used to secure the loan. It may be real estate or other investments.
Consumer Loan. It is the money lent to an individual for personal or family purposes.
Term of the loan. It refers to the time to pay the entire loan.

What is It

Amortization Method. It is the method of paying a loan (principal and interest) on an installment basis, usually of equal at regular intervals.
Chattel Mortgage. It refers to a mortgage on a movable property.
Collateral. These are assets used to secure the loan. It may be real estate or other investments.
Mortgage. This is a loan secured by collateral that the borrower is obliged to pay at specific terms.
Outstanding Balance. It refers to any remaining debt at a specified time.

What's More

Identify whether the following is a consumer or business loan.

Example No. 1. Mr. Agustin plans to have a barbershop. He wants to borrow some money from the bank for him to buy equipment and furniture for the barbershop. **Solution:** Business Loan

Example No. 2. Mr. and Mrs. Craig want to borrow money from the bank to finance their son's college education. **Solution:** Consumer Loan

Example No. 3. Mr. Alonzo wants to have some improvements on their 10-year-old house. He wants to build a new room for their 13-year-old daughter. He will borrow some money from the bank to finance this plan. **Solution:** Consumer Loan

Example No. 4. Mr. Samson owns a *siomai* food cart business. He wants to put another food cart on a new mall in the other city. He decided to have a loan to establish the new business. **Solution:** Business Loan.

What I Can Do

Outstanding Balance

Recall that the outstanding balance of a loan is the amount of the loan at this time.

Note: In this case, the house itself is used as the mortgaged property. Please take note that the other way to solve this is to directly compute the mortgaged amount by multiplying the cash value of the property by the percentage of the financed amount, which in this case, 100% - 20% = 80%. Thus the amount of the loan is given by (0.80)(₱3,000,000.00)= ₱2,400.00

One method to compute the outstanding balance is to get the present value of all remaining payments. This method is called the **Prospective Method**.

We use the symbol B_k to denote the outstanding balance after k payments. In other books, ${}_pOB_k$ is used (the "p" stands for "***prospective***")

Example No. 5 Mrs. Se borrowed some money from a bank that offers an interest rate of 12% compounded monthly. His monthly amortization for 5 years is ₱11,122.22. How much is the outstanding balance after the 12th payment?

Given: R = ₱11,122.22 $i^{(12)} = 0.12$ $j = \frac{i^{(12)}}{12} = \frac{0.12}{12} = 0.01$

$k = 12$ number of payments paid

$n - k = 48$ since only 48 payments remain

Find: Present Value of 48 future payments

(Since there are 48 payments left)

Solution: $B_k = R\left[\frac{1-(1+j)^{-(n-k)}}{j}\right]$

$$B_k = ₱11{,}122.22\left[\frac{1-(1.01)^{-48}}{0.01}\right]$$

$B_k = ₱422{,}354.73$

The outstanding balance is ₱422,354.73.

Example No. 6. Mr. & Mrs. Banal purchased a house and lot worth ₱4,000,000.00. They paid a down payment of ₱800,000.00. They plan to amortize the loan of ₱3,200,000.00 by paying monthly for 20 years. The interest rate is 12% convertible monthly.

(a) How much is the monthly payment?

Given: P = ₱3,200,000.00 $i^{(12)} = 0.12$ $j = \frac{i^{(12)}}{12} = \frac{0.12}{12} = 0.01$

n = mt = (12)(20) = 240

Find: Regular Payment R

Using the formula $P = R\left[\frac{1-(1+j)^{-n}}{j}\right]$, then $R = \frac{P}{\left[\frac{1-(1+j)^{-n}}{j}\right]}$

Solution: $R = \frac{₱3{,}200{,}000.00}{\left[\frac{1-(1.01)^{-240}}{0.01}\right]} = ₱35{,}234.76$

(b) What is the total interest paid?

Given: P = ₱3,200,000.00 R = ₱35,234.76 n = 240

Find: Total Interest Paid

Solution: There are 240 payments of ₱35,234.76. The total payment is
240 X ₱35,234.76 = ₱8,456,342.40
The Principal is only ₱3,200,000.00
Interest Amount = Total Payments – Principal
Interest Amount = ₱8,456.342.40 – ₱3,200,000.00
Interest Amount = ₱5,256,342.40
The interest amount is ₱5,256,342.40.

Note: You may be surprised to learn that much of what is being paid for the interest. This is particularly true if a loan is being paid over a long period of time.

(c) What are the principal and interest components of the 51st payment?

Given: P = ₱3,200,000.00 $i^{(12)} = 0.12$ $j = \frac{i^{(12)}}{12} = \frac{0.12}{12} = 0.01$

n = mt = (12)(20) = 240

R = ₱35,234.76

Find: Principal and Interest Components of the 51st payment

Solution: **Step 1**

Get the outstanding balance after the 50th payment (the balance after the 50th payment is what the 51st payment will be for).

Since 50 payments have been paid already, there will be 190 remaining payments.

The outstanding balance after the 50th payment is:

$$B_{50} = R\left[\frac{1-(1+j)^{-190)}}{j}\right]$$

$$B_{50} = ₱35{,}234.76\left[\frac{1-(1.01)^{-190}}{0.01}\right]$$

$$B_{50} = ₱2{,}991.477.63$$

The outstanding balance is ₱2,991,447.63.

Step 2

After the 50th payment, the outstanding balance is ₱2,991.477.63.

Since the interest rate per period is $j = 0.01$, then the remaining balance of ₱2,991,447.63 will be charged an interest of (0.01)

(₱2,991,477.63)= ₱29,914.78.

The 51st payment of ₱35,234.76 will be used to pay for this interest. Thus, the interest component I_{51} OF THE 51ST PAYMENT IS ₱29,914.78.

The remaining portion of the 51st payment is the principal component, denoted by PR_{51}, is: $PR_{51} = R - I_{50} =$ ₱35,234.76 − ₱29,914.78 = ₱5,319.98

Thus, for the 51st payment, the part that goes to pay the interest is ₱29,914.78 and the part that goes to pay the principal is ₱5,319.98

Assessment

Answer the following problems.

1. A person borrowed ₱1,000,000.00 for the purchase of a car. If his monthly payment is ₱25,000.00 on an 8-year mortgage, find the total amount of interest.
2. A consumer loan worth ₱80,000.00 is to be repaid in 12 months at 8% convertible monthly. How much is the monthly payment?
3. A business loan worth ₱250,000.00 is to repay in quarterly installments for 1 year. How much is the quarterly payment if money is worth 8% converted quarterly?

What I Need to Know

Targets:
1. Illustrate and symbolize propositions. **(M11GM-11g-1)**
2. Distinguish between simple and compound propositions. **(M11GM-11g-3)**
3. Perform the different types of operations on propositions. **(M11GM-11g-4)**

What I Know

A. State the negation of the following propositions.

1.1 a: Mary received a text message from her friend.
1.2 b: $x^2 + y^2 = z^2$

B. Let $p, q,$ and r be the propositions p: "Annie has a stomachache.", q: "Annie misses the exam.", and r: "Annie receives a passing grade from the subject." Express the following in English sentences or symbols.

2.1 "Annie does not have a stomachache, yet she misses the exam."
2.2 "If Annie has a stomachache, then she misses the exam and does not receive a passing grade for the subject."

2.3 "Either Annie has a stomachache and misses the exam, or she does not miss the exam, and she receives a passing grade for the subject.

C. Suppose p is a true proposition, q is a false proposition, and r is a true proposition. Determine the truth value of the following propositions.

3.1 $(p \wedge q) \wedge r$
3.2 $p \vee (q \vee r)$
3.3 $p \rightarrow ((\sim q) \vee r)$

Lesson 7 Key Concepts of Propositional Logic: Syllogisms and Fallacies

Proposition

A proposition is a declarative sentence that is either true or false, but not both. If a proposition is true, then its truth value is true, which is denoted by T; otherwise, its truth value is false, which is denoted by F.

Propositions are usually denoted by small letters. For example, the proposition
p: Everyone should study logic may be read as p is the proposition "Everyone should study logic".

If a sequence of the proposition is considered, we denote the proposition by p_1,p_2,....

Example No. 1. Determine whether each of the following statements is a proposition or not. If it is a proposition, give its value.

p: Mindanao is an island in the Philippines

q: Find the number which divides your age.

r: My seatmate will get a perfect score in the logic exam.

s: Welcome to the Philippines!

t: 3 + 2 = 5

u: f(x)=√x/((x+1)) is a rational function

Solution: Recall that a statement to be a proposition must be a declarative sentence, and it should have a truth value of either true or false, but not both true and false at the same time.

This is a declarative sentence, and Mindanao is an island in the Philippines. Hence p is a proposition.

This is an imperative sentence, and so it is not a proposition.

The statement is a declarative sentence, but its truth value will only be known after the logic exam. Nonetheless, it can either be true or false, but not both. Hence r is a proposition.

This statement is exclamatory, and hence t is a true proposition.

The given equation is a mathematical sentence. Translated into English, the equation reads, "the sum of three and two is five", which is a declarative sentence. It is also a true statement. Hence t is a true proposition.

This is s declarative statement. Since the numerator of the function is not a polynomial function, the function f is not rational, so the statement is false. It is, therefore, a false proposition.

What's New

Compound Proposition is a proposition formed from a simpler proposition using **logical connectors** or some combination of logical connectors.

Some logical connectors involving propositions p and/or q may be expressed as follows:

- not p
- p and q
- p or q
- if p, then q

where <, > stands for some proposition. A Proposition is simple if it cannot be broken down any further into other component propositions.

- p_1: **not** r
- p_2: f **and** I **or** b
- p_3: **If** h, **then** g
- p_4: (**If** a **then** s) **and** (**if** s **then** a**)**

What is It

Example 1:

For each of the propositions in Example No. 1, determine whether it is simple or compound. If it is a compound proposition, identify the simple components.

Solutions:

The propositions p, r, t, and u are simple propositions. On the other hand, the following are compound propositions:

- p_1: **It not the case that** $\sqrt{2}$ is a rational number.
- p_2: Either logic is fun and interesting, **or** it is boring.
- p_3: **If** you study hard, **then** you will get good grades.

- p_4: **If** you are more than 60 years old, **then** you are entitled to a Senir Citizen's card, **and if** you are entitled to a Senior Citizen's card, **then** you are more than 60 years old.

Furthermore, we can determine the simple propositions that make up the propositions p_1, p_2, p_3, and p_4. We do so in the following table:

Proposition	**Simple Components**
p_1	r: $\sqrt{2}$ is a rational number
p_2	f: logic is fun i: Logic is interesting b: Logic is boring
p_3	h: You study hard g: You get good grades
p_4	a: You are more than 60 years old s: You are entitled to a Senior Citizen's card

What's More

Logical Operators

Definition: Given a proportion, its **truth table** shows all its possible truth values

Example No. 2. Since a proportion has two possible truth values, a proportion p would have the following truth table.

P
T
F

A Truth table can also be used to display various combination of the truth values of two proportions p and q. The rows of the table will correspond to each truth value combination of p and q, there will be $2^2 = 4$ rows. The truth table for propositions p and q are as follows:

p	q
T	T
T	F
F	T
F	F

Similarly, suppose p, q, and r are propositions. Then the truth table involving the given propositions has $2^3 = 8$ rows, as shown below.

P	q	r
T	T	T
T	T	F

T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

In general, a truth table involving n propositions has 2^n rows.

Definition: The **negation** of a proposition p is denoted by $\sim p$ (not p) and is defined through its truth table

p	$\sim p$
T	F
F	T

Example No. 3 State the negation of the following propositions.

- $n_1: p(x) = \frac{(x-1)}{(x+2)}$ is a polynomial function.
- n_2: 2 is an odd number.
- n_3: The *tinikling* is the most difficult dance.
- n_4: Everyone in Visayas speaks Cebuano.

Solutions:

- $\sim n_1$:I "It is not true that $p(x) = \frac{(x-1)}{(x+2)}$ is a polynomial function" or simply "$p(x) = \frac{(x-1)}{(x+2)}$It is not a polynomial function".
- $\sim n_2$: "It is not ture that 2 is an odd number" or "2 is an even number".
- $\sim n_3$: "The *tinikling* is not the most difficult dance".
- $\sim n_4$: "Not everyone in Visayas speaks Cebuano".

Definition: The **Conjunction** of the propositions p and q is denoted by p ∧ q: (p and q) and is defined through its truth table.

P	q	p ∧ q
T	T	T
T	F	F
F	T	F
F	F	F

The proposition p and q are called **conjuncts**.

The conjunction p ∧ q is true only when both conjuncts p and q are true, as shown in its truth table.

Example No. 4 Let p and q be the propositions
p: angels exists.
q: $\pi > 3$.

Express the following conjunctions in English sentences or in symbols, as the case may be.

1. $p \wedge q$
2. $p \wedge (\sim q)$
3. "Angels do not exist and $\pi > 3$".
4. "While angels do not exist, $\pi > 3$".

Solutions:

1. "Angels exist and $\pi > 3$".
2. "Angels exist and $\pi > 3$" or "Angels exists, yet $\pi > 3$".
3. $(\sim p) \wedge (\sim q)$
4. $(\sim p) \wedge q$

Definition: The **disjunction** of two propositions p and q is denoted by

p V q: (p or q) and is defined through its truth table

The propositions p and q are called **disjuncts**.

The above truth table shows us that the disjunction p V q is false only when both disjuncts p and q are false.

P	q	p V q
T	T	T
T	F	T
F	T	T
F	F	F

Example No. 5 Let p, q, and r be the following propositions:

- p: Victor has a date with Liza.
- q: Janree is sleeping.
- r: Eumir is eating.

Express the following propositions in English sentences or in symbols, as the case may be.

1) p V q
2) $q \wedge (\sim r)$
3) $p \wedge (q$ V $r)$

Solutions:

1) "Victor has a date with Liza or Janree is sleeping".
2) "Either Janree is sleeping or Eumir is not eating".
3) "Either Victor has a date with Liza, or Janree is sleeping, or Eumir is eating".

4) "Either Victor has a date with Liza or Janree is sleeping, or Eumir is eating."
5) "Either Victor has a date with Liza and Janree is sleeping, or Eumir is eating."
6) "Either Victor has a date with Liza or Janree is sleeping, and Eumir is eating."
7) "Either Victor has a date with Liza and Janree is sleeping, or Victor has a date with Liza and Eumir is eating."

Solutions:

1) (p V q) V r
2) (p $\wedge$ q) r
3) p V (q $\wedge$ r)
4) (p $\wedge$ q) V (p $\wedge$ r)

It will be shown later that p V (q V r) and (p V q) V r are logically equivalent statements, so we can write p V q V r. Likewise, it will also be shown that p $\wedge$ (q $\wedge$ r) and (p $\wedge$ q) $\wedge$ r are logically equivalent, so that we can write p $\wedge$ q $\wedge$ r.

Example No. 6 Suppose p, q, and r are the propositions defined above. Consider the scenario that one Friday night, Victor and Janree are studying for their Logic

exam. Meanwhile, Eumir just tweeted a picture of himself eating *crispy pata* and *sisig*. What is the truth value of the proposition (~p) V (q ∧ r)?

Solution: From the given scenario, it follows that p and q are also propositions, and r is a true proposition. Therefore, the conjunction q ∧ r is false. Also, ~p is true since p is false. Hence, the disjunction (~p) V (q ∧ r)is true since at least one of its disjuncts is true.

The above discussion may also be summarized in the following table:

P	q	R	$\sim p$	$q \wedge r$	$(\sim p) V (q \wedge r)$
F	F	T	T	F	T

Definition: The **conditional** of the propositions p and q is denoted by p → q: (If p, then q) and is defined through its truth table

P	q	p → q
T	T	T
T	F	F
F	T	T
F	F	T

The conditional p → q may also be read as "p implies q). The proposition p is called the **hypothesis**, while the proposition q is called the **conclusion**.

Example No. 7. Suppose that Geebee is a Grade 11 student. Consider the following conditionals.

- p_1: If Geebee is in Grade 11, then she is a senior high school student.
- p_2: If Geebee is in Grade 11, then she is working as a lawyer.
- p_3: If Geebee has a degree in computer science, then she believes in true love.

Solution:

- p_1: The hypothesis and the conclusion are true. Thus, p_1 is true (From the first row of the truth table for the conditional statement).
- p_2: While the hypothesis is true, the conclusion is not (a Grade 11 student is not qualified to be a lawyer). From the definition of the conditional (second row of its truth table), the conditional statement of p_2 is not true.
- p_3: The hypothesis is not true since Geebee is still in Grade 11. On the other hand, we cannot determine the truth value of the conclusion "she believes in true love". From the last two rows of the truth table, regardless of the truth table of the conclusion, the conditional statement is true.

Definition: the **biconditional** of propositions p and q is denoted by p ↔ q: (p if and only if q) and is defined through its truth table.

P	q	p ↔ q
T	T	T
T	F	F
F	T	F
F	F	T

The proposition may also be written as "p iff q". The propositions p and q are the **components** of the biconditional.

Example No. 8. Suppose that Geebee is a Grade 11 student. Let us now consider the following biconditionals.

- p_1: Geebee is in Grade 11if and only if she is a high school student.
- p_2: Geebee is in Grade 11 if and only if she is working as a lawyer.
- p_3: Geebee has a degree in computer science if and only if she believes in true love.

Solution:

- p_1: Again both simple components of p_1 are true. Therefore, the biconditional statement is true.
- p_2: It is true that Geebee is in Grade 11, but it is not true that Geebee is working as a lawyer. Therefore, the biconditional is not true, referring to the second row of its truth table.
- p_3: The first simple proposition, Geebee has a degree in Computer Studies, is false. The truth value of the entire biconditional depends on the truth value of the second simple component that she believes in true love. If this is true, then the biconditional is false (referring to the third row of the truth table); otherwise, the biconditional is true.

What I Can Do

Example No. 9. One day, Richard tweeted: "If I get promoted, then stop posting selfies on Facebook." Let p be the statement "Richard gets promoted", and let q be the statement "Richard stops posting selfies on Facebook". Determine whether the conditional $p \to q$ is true given the following scenarios.

Scenario A: Richard got the promotion and promptly stopped posting selfies on Facebook.

Scenario B: Richard got promoted, but then he realized that posting selfies had become a habit he cannot break easily.

Scenario C: Richard stopped posting selfies, but he did not get the promotion.

Solution:

Scenario A: Both the hypothesis and conclusion are true, so the conditional $p \to q$ is true.

Scenario B: In the scenario, the hypothesis p is true and the conclusion is q is false. By definition, the conditional $p \to q$ is false.

Scenario C: The hypothesis is false; hence regardless of the truth value of the conclusion, the conditional $p \to q$ is false. In this case, he did not break his promise since he was not promoted in the first place.

Assessment

Problem No. 1. State the negation of the following propositions.

1. c: Eden teaches high-level mathematics to college students.
2. d: James was not able to fetch his sister from school today.
3. e: The student brings at most two bags with him every school day.

Problem No. 2 Let $p, q,$ and r be the propositions p: "Annie has a stomachache.", q: "Annie misses the exam.," and r: "Annie receives a passing grade from the subject." Express the following in English sentences or symbols, as the case may be.

1. $q \rightarrow (\sim r)$
2. $\big(p \rightarrow (\sim r)\big) \vee (q \rightarrow (\sim r))$
3. $(\sim q) \leftrightarrow r$

Problem No. 3 Suppose p is a true proposition, q is a false proposition, and r is a true proposition. Determine the truth value of the following propositions.

1. $p \wedge (q \vee r)$
2. $(p \rightarrow q) \vee r$
3. $q \rightarrow (p \vee r)$
4. $(r \leftrightarrow p) \rightarrow q$
5. $(q \vee r) \wedge (p \rightarrow r)$

What I Need to Know

Targets:

1. Determine the truth value of propositions. **(M11GM-IIh-1)**
2. Illustrate the different forms of conditional propositions. **(M11GM-IIh-2)**
3. Illustrate the different types of tautologies. **(M11GM-IIi-1)**

What I Know

Determine if the statement is a tautology, contradiction, or neither.

$$(\boldsymbol{p} \rightarrow \boldsymbol{q}) \vee (\boldsymbol{p} \rightarrow \boldsymbol{q})$$

Lesson 8	Propositions

TRUTH VALUE OF A PROPOSITION

The Truth Value of a proposition is the truthfulness or falsity of the proposition denoted by **T (True)** and **F(False).** The truth value of a compound proposition for all possible cases can be shown using a truth table. It is a device used to determine when a compound statement is true or false.

What's New

$\mathbf{2^n}$ is the formula for the truth table, where n is the number of letter formula.

OPERATION ON PROPOSITIONS

A. NEGATION

The $\neg p$ or $\sim p$ is **true** if p is **false** and vice versa.

Examples: p: Euler is German

$\sim p$: Euler is not German

Truth Table

p	$\sim p$
T	F
F	T

B. CONJUNCTION (and operation)

The conjunction $p \wedge q$ is **true** if both p and q are **true.**

Truth Table

$\sim p$	$\sim p$	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

C. DISJUNCTION (or operation)

Disjunction is **false** if both statements are **false.**

Truth Table

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

What is It

Consider the statements below,

p: Martha watched the Valenzuela Live.

q: Martha answered her assignment.

r: Martha did not attend the follow-up discussion.

Write the following statements.

1. $\sim p$
2. $\sim r$
3. $p \wedge q$
4. $q \wedge r$
5. $(p \wedge \sim q) \vee \sim r$

What's More

FORMS OF CONDITIONAL PROPOSITIONS

A conditional proposition is a proposition of the form **"if p, then q"** and is represented symbolically by **p → q.** The connective **"if – then"** is called conditional connectives. The proposition **p** is called the antecedent or hypothesis or premise, and the proposition **q** is called the consequent or conclusion.

Conditional	Meaning	Symbol
Converse	if **q**,then **p**	**q → p**
Inverse	If not **p,** then not **q**	**p →~ q**
Contrapositive	If not **q,** then not **p**	**~ q →~ p**
Biconditional	**p** if and only if **q**	**q ↔ p**

What I Can Do

1. **Chloe is home and she is not at her desk, or she is sleeping.**
   Let: p: Chloe is home
   q: Chloe is at her desk
   r: Chloe is sleeping
   Thus, the statement may be written $(p \wedge \sim q) \vee r$ Involves 3 letters; p, q, and r, hence $2^n = 2^3 = 8\ rows$

Solution:

1. $\sim q$: negate q **column**
2. $p \wedge \sim q$: use conjunction between p **column** and $\sim q$ **column**
3. $(p \wedge \sim q) \vee r$: use disjunction between $(p \wedge \sim q)$ **column** and r **column**

p	q	r	$\sim q$	$p \wedge \sim q$	$(p \wedge \sim q) \vee r$
T	T	T	F	F	**T**
T	T	F	F	F	**F**
T	F	T	T	T	**T**
T	F	F	T	T	**T**
F	T	T	F	F	**T**
F	T	F	F	F	**F**
F	F	T	T	F	**T**
F	F	F	T	F	**F**

2. **Determine if the statement is a tautology, contradiction, or neither.**
   $[\sim p \wedge (p \vee q)] \rightarrow q$ Involve 2 letters: p and q, hence $2^n = 2^2 = 4\ rows$

   Solution:
   1. $\sim p$: negate p **column**

2. $p \vee q$: use disjunction between p **column** and q **column**
3. $[\sim p \wedge (p \vee q)$**:** use conjunction between $\sim p$ **column** and $p \vee q$ **column**
4. $[\sim p \wedge (p \vee q)] \rightarrow q$**:** use "if-then" statement between $[\sim p \wedge (p \vee q)]$ **column** and q **column** and test for ***tautology***

p	q	$\sim p$	$p \vee q$	$[\sim p \wedge (p \vee q)$	$[\sim p \wedge (p \vee q)] \rightarrow q$
T	T	F	T	F	**T**
T	F	F	T	F	**T**
F	T	T	T	T	**T**
F	F	T	F	F	**T**

$\therefore$ *the statement is TAUTOLOGY*

Assessment

Determine if the statement is a tautology, contradiction or neither.
$(p \vee q) \wedge [\sim (p \wedge q)]$

What I Need to Know

Targets:

1) Illustrates different types of tautologies and fallacies. **(M11GM-Iii-1)**
2) Determine the validity of categorical syllogism. **(M11GM-Iii-2)**
3) Establishes the validity and falsity of real-life arguments using logical propositions, syllogisms, and fallacies. **(M11GM-Iii-3)**

What I Know

Write T if the statement is true and F if false.

Determine whether the given is a valid argument or fallacy.

1) Either Alvin sings or dances with Nina. Alvin sang with Nina. Therefore, Alvin did not dance with Nina.
2) Either Alvin sings or dances with Nina. Alvin did not dance with Nina. Therefore, Alvin sang with Nina.
3) It is not true that Alvin sings and dances with Nina. Alvin did not sing with Nina. Therefore, Alvin danced with Nina.

Lesson 9 Key Methods of Proof and Disproof

What's New

An **Argument** is a compound proposition of the form $(p_1 \wedge p_2 \wedge \ldots \wedge p_n) \rightarrow q$. The propositions $p_1, p_2, \ldots, p_n$ are the **premises** of the argument, and q is the conclusion. Arguments can be written in **propositional form**, as in above, or in column or **standard form**:

$$\begin{array}{c} p_1 \\ p_2 \\ \vdots \\ \underline{p_n} \\ q \end{array}$$

An argument is valid if it satisfies the validity condition: Is it logically impossible for the premises to be true and the conclusion is false? If the answer is affirmative, we say that the argument satisfies the validity condition, and is hence valid.

For argument A, we ask: can $p \rightarrow q$ and q be both true and q be false? To answer this, we look at the truth table for $p \rightarrow q$:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

The highlighted row shows that both p and $p \rightarrow q$ are true if and only if q is true. Based on the truth table, it is not possible for p and $p \rightarrow q$ to be true and q to be false. Hence, argument A is valid.

For argument A',we ask: can $p \rightarrow q$ and q be both true and p be false? Looking the same truth table,

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

The highlighted row shows that it is possible for $p \rightarrow q$ and q be both true and p be false. Hence, argument A' does not satisfy the validity condition and so it is not a valid argument.

A **valid argument** satisfies the validity condition; that is, the conclusion q is true whenever the premises $p_1, p_2, \ldots, p_n$ are all true. Alternatively, the argument is valid if the conditional $(p_1 \wedge p_2 \wedge \ldots \wedge p_n) \rightarrow q$ is tautology.

Six Rules of Syllogism

Syllogism has three main categories, thus, it also requires three components such as: The Major Premise, The Minor Premise, and The Conclusion to make a strong and accurate argument.

1. There must be THREE TERMS. The major premise, The minor premise, and the conclusion. No more, no less.
2. The minor premise must be distributed in at the least one other premise.
3. Any terms distributed in the conclusion must be distributed in the relevant premise.
4. Do not use TWO NEGATIVE premises.
5. If one of the two premises is negative, the conclusion must be negative.
6. From two universal premise, no conclusion may be drawn.

Rules of Inference	**Propositional Form**	**Standard Form**
Rule of Simplification	$(p \wedge q) \rightarrow p$	$\begin{array}{l} p \wedge q \\ \hline \therefore\ p \end{array}$
Rule of Addition	$p \rightarrow (p \vee q)$	$\begin{array}{l} p \\ \hline \therefore\ p \vee q \end{array}$
Rule of Conjunction	$(p \wedge q) \rightarrow (p \wedge q)$	$\begin{array}{l} p \\ q \\ \hline \therefore\ p \wedge q \end{array}$
Modus Ponens	$[(p \rightarrow q)] \wedge p \rightarrow q$	$\begin{array}{l} p \rightarrow q \\ p \\ \hline \therefore\ q \end{array}$
Modus Tollens	$[(p \rightarrow q) \wedge (\sim q)] \rightarrow (\sim p)$	$\begin{array}{l} p \rightarrow q \\ \sim q \\ \hline \therefore\ \sim p \end{array}$
Law of Syllogism	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$	$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore\ p \rightarrow r \end{array}$
Rule of Disjunctive Syllogism	$[(p \vee q) \wedge (\sim p)] \rightarrow q$	$\begin{array}{l} p \vee q \\ \sim p \\ \hline \therefore\ q \end{array}$
Rule of Contradiction	$[(\sim p) \rightarrow \emptyset] \rightarrow p$	$\begin{array}{l} (\sim p) \rightarrow \emptyset \\ \hline \therefore\ p \end{array}$
Rule of Proof by Cases	$[(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow [(p \vee q) \rightarrow r]$	$\begin{array}{l} p \rightarrow r \\ q \rightarrow r \\ \hline \therefore\ (p \vee q) \rightarrow r \end{array}$

Common Fallacies	**Propositional Form**	**Standard Form**
Fallacy of the Converse	$[(p \rightarrow q) \wedge \mathrm{q}] \rightarrow p$	$\begin{array}{l} p \rightarrow q \\ q \\ \hline \therefore\ p \end{array}$
Fallacy of the Inverse	$[(p \rightarrow q) \wedge (\sim \mathrm{p})] \rightarrow (\sim q)$	$\begin{array}{l} p \rightarrow q \\ \sim p \\ \hline \therefore\ \sim q \end{array}$
Affirming the Disjunct	$[(p \vee q) \wedge \mathrm{p}] \rightarrow (\sim q)$	$\begin{array}{l} p \vee q \\ p \\ \hline \therefore\ \sim q \end{array}$

Common Fallacies	**Propositional Form**	**Standard Form**
Fallacy of the Consequent	$(p \to q) \to (q \to p)$	$\therefore \dfrac{p \to q}{q \to p}$
Denying a Conjunct	$[\sim(p \wedge q) \wedge (\sim p)] \to q$	$\therefore \dfrac{\sim(p \wedge q) \quad \sim p}{q}$
Improper Transportation	$(p \to q) \to [(\sim p) \to (\sim q)]$	$\therefore \dfrac{p \to q}{(\sim p) \to (\sim q)}$

What's More

Sample Problems:

1. **Categorical Syllogism**
   All cars have wheels. I drive a car. Therefore, my car has wheels.
   Major Premise: All cars have wheels.
   Minor Premise: I drive a car.
   Conclusion: My car has wheels.

2. **Conditional Syllogism**
   If Katie is smart, then she will get into a good college
   Major Premise: Katie is smart.
   Minor Premise: Because she is smart, Katie will get good grades.
   Conclusion: If Katie is smart, then she will get into college.

3. **Syllogistic Fallacy**
   All crows are black. The bird in my cage is black. Therefore, this bird is a crow.
   Major Premise: All crows are black.
   Minor Premise: The bird in my cage is black.
   Conclusion: This bird is a crow.

What I Can Do

Determine the following whether the given is valid argument or a fallacy.

1. All laptops are tablets. Some tablets are mobiles. Therefore, mobiles are televisions.
2. All cars have wheels. I drive a car. Therefore, my car has wheels.
3. All insects frighten me. That is an insect. Therefore, I am frightened.
4. If Katie is smart, then she will get into a good college.
5. This cake is either red velvet or chocolate. Since it's not chocolate cake, it must be red velvet.

Assessment

Determine whether the following arguments are valid. If it is valid, then identify the rule of inference which justifies its validity. Otherwise, state a counter example or identify the type of fallacy exhibited by the argument.

1. If x is an odd integer, then $2x + 1$ is also an odd integer. If $2x + 1$ is an odd integer, then $3(2x + 1)$ is an odd integer. Therefore if x is an odd integer, then $3(2x + 1)$ is an odd integer.

2. If Jason has PhD and has done a considerable amount of research, then he is qualified for the research professor position. He was informed that he is qualified for the research professor position. Therefore, Jason has a PhD and has done a considerable amount of research.

Additional Activities

Determine the following whether the given is valid argument or a fallacy.

1. All crows are black. The bird in my cage is black. Therefore, this bird is a crow.
2. The scenery in Valenzuela City is beautiful. I'm in Valenzuela City. Therefore, the scenery must be beautiful.
3. On the TV show XX, Claire's husband is either dead or alive. Since he's not alive, he must be dead.

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